**BUSINESS ANALYTICS**

**POWER BI PROJECT REPORT**

**Experiment No: 1**

**Methodology / Steps Followed**

**Step 1: Data Collection**

The required datasets were collected from different data sources.

- Sales data was obtained from Excel files.
- Additional data was obtained from CSV files.
- Product and other business-related data were stored in the MySQL database.
- The required tables were identified for analysis and reporting.

**Step 2: Importing Data into Power BI**

Power BI Desktop was opened, and the datasets were imported using the **Get Data** option.

The following connections were performed:

- Connected to Excel files and imported the required sales data.
- Connected to CSV files and imported the available datasets.
- Connected Power BI to the MySQL database.
- Selected the required MySQL database and tables.
- Loaded the required data into Power BI for further processing.

**Step 3: Data Cleaning and Transformation**

The imported data was cleaned and transformed using **Power Query Editor**.

The following operations were performed:

- Removed unnecessary columns.
- Checked and removed duplicate records.
- Handled missing or null values.
- Renamed columns appropriately.
- Corrected the data types of columns.
- Converted date columns into the required date format.
- Checked numerical columns such as sales amount and quantity.
- Created a clean and structured dataset for analysis.

**Step 4: Data Modelling**

After cleaning the data, the required data model was created.

The following operations were performed:

- Created relationships between the available tables.
- Connected sales-related data with product and store-related information.
- Verified the relationships between tables.
- Used appropriate one-to-many relationships where required.
- Ensured that the data model supported accurate reporting and analysis.

**Step 5: Creating Calculated Columns and Measures**

DAX calculations were used to generate important business measures.

The important calculations included:

- Total Sales
- Total Orders
- Total Quantity Sold
- Average Sales
- Sales comparison with the previous period/month
- Sales by Store Location
- Sales by Product Category
- Weekday Sales
- Weekend Sales
- Sales by Hour
- Sales by Day of the Week

These measures were used to display the business performance in the Power BI dashboard.

**Step 6: Dashboard Development**

An interactive **Coffee Shop Sales Report** dashboard was created in Power BI.

The dashboard includes the following KPI cards:

- **Total Sales: $157K**
- **Total Orders: 3527**
- **Total Quantity Sold: 48,233**

These KPI cards provide a quick overview of the overall business performance.

**Step 7: Operations Performed and Visualizations Created**

The following operations and visualizations were performed in the Power BI report:

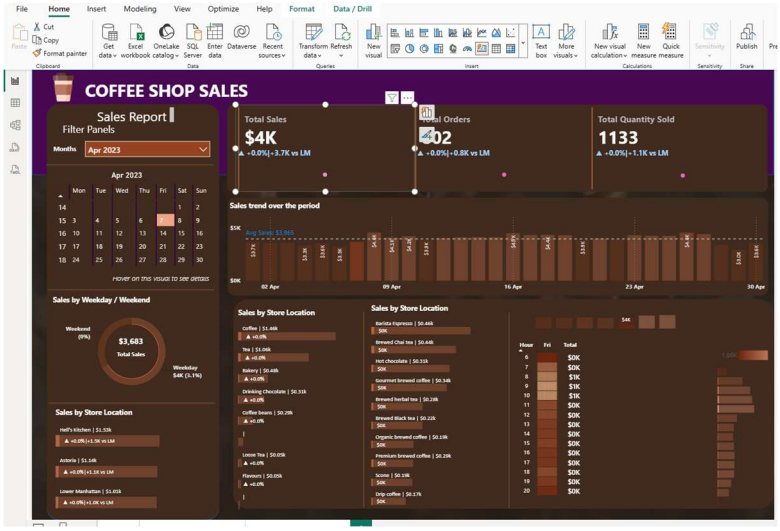
**1. Filter Panel**

A filter panel was created to allow users to interact with the dashboard.

The filters include:

- Month selection
- Date selection using a calendar
- Interactive filtering of dashboard visuals

For example, the dashboard displays data for **May 2023**.




**2. KPI Cards**

Three KPI cards were created to display:

- Total Sales
- Total Orders
- Total Quantity Sold

The KPI cards also show the comparison with the previous period to understand business growth or decline.

**3. Sales Trend Over the Period**

A column chart was created to display the sales trend across different dates.

This visualization helps in:

- Identifying daily sales performance.
- Comparing sales between different days.
- Identifying high-sales and low-sales periods.
- Understanding the overall sales trend during the selected month.

**4. Sales by Weekday / Weekend**

A donut chart was created to compare sales generated during:

- Weekdays
- Weekends

This analysis helps the management understand whether the business generates more revenue during working days or weekends.

**5. Sales by Store Location**

Bar charts were created to compare sales performance across different store locations.

This visualization helps in:

- Identifying the best-performing store.
- Comparing sales between store locations.
- Identifying low-performing locations.
- Supporting location-based business decisions.

**6. Product-Wise Sales Analysis**

A bar chart was created to analyze the sales performance of different products.

The dashboard displays products such as:

- Coffee
- Tea
- Bakery items
- Drinking Chocolate
- Coffee Beans
- Packaged Chocolate
- Other coffee shop products

This helps identify the products that contribute the most to total sales.

**7. Hour and Day Sales Analysis**

A heatmap-style visualization was created to analyze sales based on:

- Hour of the day
- Day of the week

This visualization helps identify:

- Peak business hours.
- Days with higher customer activity.
- Low-sales hours.
- Important time periods for improving store operations.

**8. Interactive Report Design**

The dashboard was designed to be interactive by using:

- Filters
- Slicers
- KPI cards
- Bar charts
- Column charts
- Donut charts
- Heatmap analysis
- Store location analysis
- Product performance analysis

All visuals are connected so that the user can interact with the report and obtain different business insights.

**Output**

The final Power BI dashboard successfully provides an interactive overview of the Coffee Shop Sales business.

The dashboard provides:

- Total Sales Performance
- Total Orders Analysis
- Total Quantity Sold
- Daily Sales Trends
- Weekday and Weekend Sales Comparison
- Store Location Performance
- Product-Wise Sales Analysis
- Hour-Wise Sales Analysis
- Day-Wise Sales Analysis
- Monthly and Date-Based Filtering
- Interactive Business Insights

The dashboard provides a centralized view of the business data and helps management monitor overall performance.

**Result**

The required business datasets were successfully imported from multiple sources, including Excel/CSV files and MySQL, into Power BI. The data was cleaned, transformed, and organized using Power Query and the required data relationships were established.

An interactive Coffee Shop Sales Dashboard was successfully developed. The dashboard displays important KPIs such as Total Sales, Total Orders, and Total Quantity Sold, along with sales trends, store performance, product analysis, weekday/weekend analysis, and hourly sales patterns.

**Conclusion**

This experiment successfully demonstrated how business data from multiple sources can be integrated into Power BI for centralized reporting and analysis. The data was imported from Excel/CSV files and MySQL, followed by data cleaning and transformation using Power Query.

Data modeling and DAX calculations were used to generate meaningful measures and business insights. The final Coffee Shop Sales Dashboard provides an interactive view of important business metrics such as sales, orders, quantity sold, store performance, product performance, and sales trends.

The developed dashboard helps management understand business performance, identify high-performing products and locations, analyze customer activity patterns, and make effective data-driven decisions.

**Learning Outcomes**

Through this experiment, I learned:

- How to import data from Excel files into Power BI.
- How to import data from CSV files into Power BI.
- How to connect Power BI with a MySQL database.
- How to select and load MySQL tables into Power BI.
- How to clean and transform data using Power Query.
- How to remove duplicate records and handle missing values.
- How to change and manage data types.
- How to create relationships between multiple tables.
- How to create DAX measures and calculations.
- How to create KPI cards for business reporting.
- How to create bar charts, column charts, and donut charts.
- How to create interactive filters and slicers.
- How to analyse sales by date, hour, weekday, and weekend.
- How to analyse store location and product performance.
- How to design an interactive and professional Power BI dashboard.

- How to generate meaningful business insights from raw data.

GitHub link : https://github.com/Amaraiah11/Amaraiah_PowerBI_Project